

# MATH

OR CBSE BOARD PATTERN

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1. (4)
2. (3)
3. (4)
4. (2)
5. (2)
6. (2)
7. (3)
8. (2)
9. (4)
10. (4)
11. (3)
12. (4)
13. (3)
14. (2)
15. (4)
16. (1)
17. (3)
18. (2)
19. (3)
20. (2)

$$4x + 5 = 22$$

$$4x - 2y = 16$$

$$1 = 1$$

$$x - 2y = 8 \quad D_1 = 4 \times 5 = 20$$

$$-3x + 4y = 0 \quad D_2 = 1$$

$$1 = -2 \times 5 = -10$$

$$\cos \theta = \frac{1}{\sqrt{3}} \quad \sin \theta = \frac{45}{5} = 1$$

$$46 = 1 \times 16 = 16$$

$$\sin \theta = \frac{1}{\sqrt{3}} \quad x + 46 = 16$$

$$x + 46 = 16$$

$$\sec 60^\circ + 10 \sec 30^\circ = 1$$

$$\sqrt{3} = \sqrt{3}$$

$$2 + \frac{2}{\sqrt{3}} = \frac{4\sqrt{3} + 2}{\sqrt{3}}$$

$$\frac{4}{\sqrt{3}} = \frac{4\sqrt{3}}{3}$$

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21. Let us assume that  $5+2\sqrt{3}$  is rational  
 $5+2\sqrt{3} = \frac{a}{b}$  (a, b are coprime)

$$2\sqrt{3} = \frac{a-5}{b}$$

$$\sqrt{3} = \frac{a-5b}{2b}$$

$$\sqrt{3} = \frac{a-5b}{2b}$$

$\Rightarrow \frac{a-5b}{2b}$  is rational

$\Rightarrow \sqrt{3}$  is irrational

$\Rightarrow$  It is a contradiction

$\Rightarrow$  My assumption is wrong  
 $\therefore 5+2\sqrt{3}$  is irrational

22. Given

$$CP = OD \quad \text{--- (i)}$$

PS is arc

$$OS = OR \quad \text{--- (ii)}$$

Apply (i) by (ii)

$$OS = OR$$

$$CP = OD$$

In  $\triangle OCP$  &  $\triangle ODR$

$$OP = OR \quad (OA)$$

$$OS = OR$$

$$CP = OD$$

by

A

Hence

23. Given

OP is tangent

only 4 can be

Because only

$$24. \frac{cos \theta - sin \theta}{cos \theta + sin \theta}$$

$$\Rightarrow \frac{cos \theta - sin \theta}{cos \theta + sin \theta}$$

$$\Rightarrow \frac{cos \theta - sin \theta}{cos \theta + sin \theta}$$

$$\Rightarrow \frac{cos \theta - sin \theta}{cos \theta + sin \theta}$$

$$\Rightarrow \frac{cos \theta - sin \theta}{cos \theta + sin \theta}$$

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$$\Rightarrow \frac{cos \theta - sin \theta}{cos \theta + sin \theta}$$

by SAS  
 $\triangle OBP \sim \triangle OPA$   
 Hence statement true

23.  $63.43^\circ$   
 It is tangent at point P



only 1 circle  
 can be drawn.

Because for a tangent at a point  
 only one circle can be drawn

24.  $\cos \theta - \sin \theta = 1/\sqrt{2}$   
 $\tan \theta = \sin \theta$

$$\Rightarrow \frac{\cos \theta - \sin \theta}{\cos \theta} = \frac{1/\sqrt{2}}{1/\sqrt{2}}$$

$$\frac{\cos \theta - \sin \theta}{\cos \theta}$$

$$\Rightarrow \frac{1 - \tan \theta}{1 + \tan \theta} = \frac{1 - \sqrt{3}}{1 + \sqrt{3}}$$

or comparing

$$\tan \theta = \sqrt{3} = \tan 60^\circ$$

is  $\theta = 60^\circ$

25.

area

Given

$$\text{area of semi-circle} = 30800 \text{ cm}^2$$

$$\text{Perimeter of circle} = \pi r + 2r$$

A/c to Q

$$\frac{\pi r^2}{2} = 30800$$

$$\Rightarrow r^2 = \frac{30800 \times 2 \times 7}{22}$$

$$\Rightarrow r^2 = 1400 \times 2 \times 7$$

$$\Rightarrow r^2 = 19600$$

$$\Rightarrow r = 140 \text{ m}$$

Perimeter of semi-circle field

$$= \pi r + 2r$$

$$\Rightarrow \frac{22}{7} \times 140 + 2 \times 140$$

$$= 440 + 280$$

$$\Rightarrow 720 \text{ m}$$

$$\therefore \text{Perimeter of field} = 720 \text{ m}$$

26.

Given

$$\text{LCM}(6^4, 8^2, 12^3) = 12^4$$

$$6^4 = 1296$$

$$8^2 = 64$$

$$12^3 = 20736$$

27.

$$\begin{array}{r}
 2 \overline{) 1296} \\
 2 \overline{) 648} \\
 2 \overline{) 324} \\
 2 \overline{) 162} \\
 2 \overline{) 81} \\
 3 \overline{) 27} \\
 3 \overline{) 9} \\
 3 \overline{) 3}
 \end{array}$$

$$\therefore 6^4 = 2^4 \times 3^4$$

$$8^k = 2^k$$

$$\therefore \text{LCM of } (6^4, 8^k) = 12^k$$

$$\Rightarrow 2^6 \times 3^4 \times 2^k = 12^k \text{ [given]}$$

$$k = 12^k$$

$$k = \frac{2^6 \times 3^4}{2^k \times 3^k}$$

$$k = 64 \times 2^k$$

$$\boxed{k=4}$$

Smallest value of  $k=4$

$$27. \quad x^2 - 6x + 3(k-1) = 0$$

$\alpha, \beta$  are roots

$$\alpha + \beta = -(-6) = 6 \quad \text{--- (1)}$$

$$\alpha - \beta = 1 \text{ [given]} \quad \text{--- (2)}$$

$$\text{(1) + (2)}$$

$$\alpha + \beta = 6$$

$$\alpha - \beta = 1$$

$$\underline{2\alpha = 7}$$

$$\alpha = 3$$

Sub in ①

$$\beta = 5 - 3$$

$$\beta = 2$$

$$d\beta = 3(k-1)$$

$$(3)(2) = 3(k-1)$$

$$6 = 3(k-1)$$

$$2 = k-1$$

$$\boxed{k = 3}$$

28. let fraction =  $\frac{x}{y}$

A/c to Q

$$\frac{x-2}{y} = \frac{1}{3}$$

$$y = 3x - 6$$

$$3x - y - 6 = 0 \quad \text{--- (1)}$$

$$\frac{x}{y+1} = \frac{1}{2}$$

$$y+1 = 2x$$

$$2x - y + 1 = 0 \quad \text{--- (2)}$$

$$\text{①} + \text{②}$$

29. (a)

As



$$3x - y - 6 = 0$$

$$2x - y + 1 = 0$$

$$x - 7 = 0$$

$$x = 7$$

sub in ①

$$3(7) - y = 6$$

$$-y = 6 - 21$$

$$-y = -15$$

$$y = 15$$

$$\therefore \text{fraction} = \frac{7}{15}$$

29 (a)

As given,

radius = 28 cm

ABC is quadrant

$$\therefore \angle BAC = 90^\circ$$

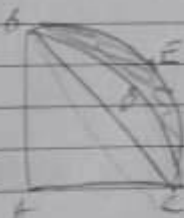
In  $\triangle ABC$

$$BC^2 = AC^2 + AB^2$$

$$= (28)^2 + (28)^2$$

$$BC^2 = 2(28)^2$$

$$BC = 28\sqrt{2} \text{ cm}$$



$$\text{radius of semi-circle} = \frac{28\sqrt{2}}{2}$$

$$= 14\sqrt{2} \text{ cm}$$

Area of semi-circle =  ~~$\frac{1}{2} \times \pi \times r^2$~~

$$\frac{\pi r^2}{2}$$

$$\Rightarrow \frac{22}{7} \times \frac{1}{2} \times (14.5)^2$$

$$\Rightarrow \frac{22}{7} \times \frac{1}{2} \times 14.5 \times 14.5$$

$$\Rightarrow 616 \text{ cm}^2$$

Area of triangle =  $\frac{1}{2} r^2 \sin \theta$

$$= \frac{1}{2} \times 28 \times 28 \times \sin 90$$

$$= \frac{1}{2} \times 28 \times 28 \times 1$$

$$= 392 \text{ cm}^2$$

Area of shaded region =  $\frac{1}{4} \pi r^2$

$$= \frac{1}{4} \times 22 \times 28 \times 28$$

$$= 616 \text{ cm}^2$$

Area of shaded region = Area of semi-circle + Area of the shaded quadrant

$$\Rightarrow 616 + 392 - 616$$

$$= 392 \text{ cm}^2$$



1) Given

area of circular playground =  $22176 \text{ cm}^2$

$$\pi r^2 = 22176 \text{ cm}^2$$

$$\Rightarrow 22 \times r^2 = 22176$$

$\Rightarrow$

$$r^2 = \frac{22176 \times 7}{22 \times 7}$$

$$r^2 = 1008 \times 7$$

$$r^2 = 7056$$

$$r = 84 \text{ m}$$

Circumference of circle =  $2\pi r$

$$= 2 \times \frac{22}{7} \times 84$$

$$\Rightarrow 44 \times 12$$

$$\Rightarrow 528 \text{ m}$$

$$\therefore \text{cost of fencing} = 528 \times 50$$

$$= 26400/-$$

30  $\frac{\tan \theta + \cot \theta}{1 - \cot \theta} = 1 + \tan \theta + \cot \theta$

$$\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = \frac{1 - \cos \theta}{\sin \theta} + \frac{1 - \sin \theta}{\cos \theta}$$

$$\Rightarrow \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$$

$$\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}$$

$$\Rightarrow \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta (\sin \theta + \cos \theta) \sin \theta (\cos \theta - \sin \theta)}$$

$$\Rightarrow \frac{\sin^2 \theta - \cos^2 \theta}{\cos \theta (-\sin \theta - \cos \theta) \sin \theta (\sin \theta - \cos \theta)}$$

$$\Rightarrow \frac{\sin^2 \theta - \cos^2 \theta}{\sin \theta \cos \theta (\sin \theta - \cos \theta)}$$

$$\Rightarrow \frac{(\sin \theta - \cos \theta)(\sin^2 \theta + \cos^2 \theta + \sin \theta \cos \theta)}{\sin \theta \cos \theta (\sin \theta - \cos \theta)}$$

$$\Rightarrow \frac{\sin^2 \theta + \cos^2 \theta + \sin \theta \cos \theta}{\sin \theta \cos \theta}$$

$$\Rightarrow \frac{\sin^2 \theta}{\sin \theta \cos \theta} + \frac{\cos^2 \theta}{\sin \theta \cos \theta} + \frac{\sin \theta \cos \theta}{\sin \theta \cos \theta}$$

$$\Rightarrow \tan \theta + \cot \theta + 1$$

$\Rightarrow$  RHS

$$\text{LHS} = \text{RHS}$$

Hence proved

31.

3) Given, median = 16  
total frequency = 70

| class | f  | CF     |
|-------|----|--------|
| 0-5   | 12 | 12     |
| 5-10  | a  | 12+a   |
| 10-15 | 12 | 24+a   |
| 15-20 | 15 | 39+a   |
| 20-25 | b  | 39+a+b |
| 25-30 | 6  | 45+a+b |
| 30-35 | 6  | 51+a+b |
| 35-40 | 4  | 55+a+b |
|       | 70 |        |

Total  $N = 70$   $\therefore \frac{N}{2} =$

$55 + a + b = 70$

$a + b = 15 \quad \text{--- (1)}$

median class = 15-20

$L = 15$ ,  $cf = 24 + a$ ,  $n_2 = 35$ ,  $h = 5$

median =  $L + \frac{(n_2 - cf)}{f} \times h$

$16 = 15 + \frac{(35 - (24 + a)) \times 5}{15}$

$16 = 15 + \frac{35 - 24 - a}{3}$

$3 = 11 - a$

$a = 8$

Sub in (1)

$$b = 75 - a$$

$$b = 7$$

$$\therefore a = 8, b = 7$$

32. Let no. of books brought =  $x$   
cost = 250

At 200 = 0

(each 40)

If no. of books brought =  $(x+4)$   
then cost =  $\frac{80}{x+4}$

A/C to Q

$$\frac{80}{x} - \frac{80}{x+4} = 1$$

$$\Rightarrow \frac{80(x+4) - 80(x)}{x(x+4)} = 1$$

$$\Rightarrow \frac{80x + 320 - 80x}{x^2 + 4x} = 1$$

$$\Rightarrow \frac{320}{x^2 + 4x} = 1$$

$$\Rightarrow x^2 + 4x = 320$$

$$\Rightarrow x^2 + 4x - 320 = 0$$

$$\Rightarrow x^2 + 20x - 16x - 320 = 0$$

$$\Rightarrow x(x+20) - 16(x+20) = 0$$

$$(x+20)(x-16) = 0$$

$$\therefore x = -20 \text{ or } x = +16$$

not possible

Ex. 17.16

∴ He brought 16 birds initially.

Ex. 22.45.  
 $AD = DC$   
 $OD = OX$



BE & AD meet each other &  $ABXC$  is a parallelogram

- ∴  $AX \parallel OC$  ;  $AC \parallel BX$
- ∴  $AX \parallel OC$  &  $CX \parallel BO$
- ∴  $AX \parallel OC$  &  $CX \parallel BO$

In  $\triangle ABX$ ,  $AX \parallel OC$

$$\frac{AO}{AX} = \frac{AE}{AB} \quad \text{--- (1)}$$

$$\Rightarrow \frac{AO}{AX} = \frac{AE}{AB} \quad \text{--- (1) (proved)}$$

In  $\triangle ACX$ ,  $CX \parallel BO$

$$\frac{AO}{AX} = \frac{AE}{AC} \quad \text{--- (2)}$$

from (1) & (2)

$$\frac{AE}{AB} = \frac{AE}{AC}$$

∴ (1)  $EF \parallel BC$  (converse of BPT)

(1)  $EF \parallel BC$  (proved)

(2)  $AO : AX = AE : AB$  (proved)

34. Given dimensions:  
850cm, 625cm & 475cm

length of longest rod = HCF(850, 625, 475)

|    |     |   |     |    |    |
|----|-----|---|-----|----|----|
| 5  | 170 | 5 | 125 | 5  | 95 |
| 5  | 34  | 5 | 25  | 19 | 19 |
| 17 | 2   | 5 | 5   |    |    |
|    | 1   |   | 1   |    |    |

$$850 = 5^2 \times 2 \times 17$$

$$625 = 5^4$$

$$475 = 5^2 \times 19$$

$$\text{HCF} = 5^2$$

$$= 25$$

length of longest rod = 25cm

35. Given

$$a = 3, a_n = 83, S_n = 903$$

$$S_n = \frac{n}{2} (a + a_n)$$

$$903 = \frac{n}{2} (3 + 83)$$

$$903 = \frac{n}{2} (86)$$

$$903 = 43n$$

36. Given

$$a =$$

(i)

(ii)



$$n = 203$$

$$\frac{48}{5}$$

$$[n-2]$$

$$\text{no. of terms} = 21$$

$$a_n = a + (n-1)d$$

$$23 = 3 + (21-1)d$$

$$20 = 20d$$

$$20 = 20d$$

$$d = 1$$

$$\Rightarrow$$

$$\text{div } (d=4)$$

$$\therefore \begin{cases} \text{no. of terms} = 21 \\ d = 4 \end{cases}$$

36.

Given,

$$a = 1000, \quad d = 100, \quad a_n = 1,18,000$$

$$\begin{aligned} \text{(i)} \quad a_{10} &= a + 9d \\ &= 1000 + 9(100) \\ &= 1000 + 900 \\ a_{10} &= 1900 \end{aligned}$$

$$\therefore \text{amt paid in 10th installment} = 1900/-$$

$$\begin{aligned} \text{(ii)} \quad a_{40} &= a + 39d \\ &= 1000 + 39(100) = 1000 + 3900 \\ &= 4900 \end{aligned}$$

$$\text{amt paid in last installment} = 4900/-$$

(ii)  $a = 1000$ ,  $d_{10} = 4900$   
 $a : a_{10}$   
 $1000 : 4900$   
 $10 : 49$

37. Given  
 $AP = 31, 27, 23, \dots, 3$   
 $a = 31$ ,  ~~$a = 27$~~   $d = -4$

(i)  $a_n = a + (n-1)d$

(ii)  $a_5 = a + 4d$   
 $= 31 + 4(-4)$   
 $= 31 - 16$   
 $= 15$   
 no. of rows is  $5^{th} row = 15$

(iii)  $a_n = 11$   
 $a_n = a + (n-1)d$   
 $11 = 31 + (n-1)(-4)$   
 $11 = 31 - 4n + 4$   
 $11 = 35 - 4n$   
 $35 - 11 = 4n$   
 $24 = 4n$   
 $n = \frac{24}{4}$   
 $n = 6$   
 $6^{th} row$  has 11 rows

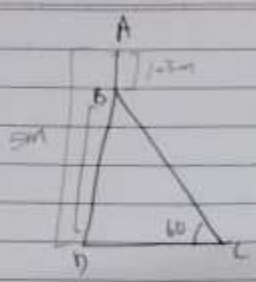
38. Given  
 pole h  
 AB =

(9) BD

(ii)

(iii)

38. Given,  
 pole height (AD) = 5m  
 AB = 1.3m



$\therefore$   
 (i)  $BD = AD - AB$   
 $= 5 - 1.3$   
 $BD = 3.7m$

(ii) In  $\triangle BDC$

|                                       |                                       |
|---------------------------------------|---------------------------------------|
| $\frac{BD}{DC} = \sin 60^\circ$       | $\sin 60^\circ = \frac{BD}{BC}$       |
| $\frac{\sqrt{3}}{2} = \frac{3.7}{BC}$ | $\frac{\sqrt{3}}{2} = \frac{3.7}{BC}$ |
| $BC = \frac{3.7 \times 2}{\sqrt{3}}$  | $BC = \frac{7.4}{\sqrt{3}}$           |
| $BC = \frac{7.4 \times \sqrt{3}}{3}$  | $BC = \frac{7.4\sqrt{3}}{3}$          |

$\therefore$  length of ladder =  $\frac{7.4\sqrt{3}}{3}m$

(iii) If  $\theta = 30^\circ$ , then

$\sin 30^\circ = \frac{BD}{BC}$   
 $\frac{1}{2} = \frac{3.7}{BC}$   
 $BC = 7.4m$

$\therefore$  length of ladder =  $7.4m$